Maximum Sum of 3 Non-Overlapping Subarrays (hard)

Problem Statement

Given an array of integers `nums` and an integer `k`, find the starting indices of three non-overlapping subarrays having `maximum sum` and each of a length `k`.

Return the array of size 3 containing the starting indices of such subarrays. If `multiple answers` exist, return the `lexicographically smallest` answer.

Examples

Example 1:

- **Input**: `nums = [1, 2, 3, 1, 1, 2, 1, 3, 2], k = 2`
- **Expected Output**: `[1, 4, 7]`
- **Justification**: The subarrays would be `[2, 3]`, `[1, 2]`, and `[3, 2]` with sums 5, 3, and 5 respectively.

Example 2:

- **Input**: `nums = [4, 5, 10, 6, 11, 17, 4, 5, 6, 7], k = 3`
- **Expected Output**: `[0, 3, 7]`
- **Justification**: The subarrays would be `[4, 5, 10]`, `[6, 11, 17]`, and `[5, 6, 7]` with sums 19, 34, and 18 respectively.

Example 3:

- **Input**: `nums = [1, 1, 1, 1, 1, 1, 1, 1, 1], k = 2`
- **Expected Output**: `[0, 2, 4]`
- **Justification**: The subarrays would be `[1, 1]`, `[1, 1]`, and `[1, 1]` with each having a sum of 2.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Arrays;

public class Solution {
    public int[] maxSumOfThreeSubarrays(int[] nums, int k) {
        int[] ans = new int[3];
        // ToDo: Write Your Code Here.
        return ans;
    }
}
```

Test Case | Run Code Result | Stdout

```
[1,2,1,2,6,7,5,1]
2
```

Use Example Testcases | Run | Submit

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Solution: Maximum Sum Of 3 Non-Overlapping Subarrays

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